

Rallis inner product formula

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1 Overview

We set up the notation for this note. Let F be a number field and E/F is a quadratic extension. We denote by $\mathbb{A} = \mathbb{A}_F$ the ring of adèle of F . Let G be a connected reductive group with trivial split center over F . We write $G(\mathbb{A})$ for the adèle points of G and write $[G]$ for the automorphic quotient.

We have the following space of functions on $[G]$:

$$\mathcal{A}_{\text{cusp}}(G) \subset \mathcal{A}_{\text{disc}}(G) \subset \mathcal{A}(G).$$

Such that

$$\mathcal{A}_{\text{cusp}}(G) \subset \mathcal{S}([G]), \quad \mathcal{A}(G) \subset \mathcal{T}([G]) \subset C^\infty([G]),$$

where $\mathcal{S}([G])$ is the space of Schwartz function on $[G]$ and $\mathcal{T}([G])$ is the space of uniform moderate growth functions on $[G]$.

Fix $\varepsilon \in \{\pm 1\}$. Let V be an E/F ε -Hermitian space over E , we will denote by $H := H(V)$ the unitary group of V . Let U be a E/F $-\varepsilon$ -Hermitian space, we will denote by $G = G(U)$ the unitary group of U . Let $W := U \oplus U^-$ (doubled “ U ”), it is a $2n$ -dimensional split $-\varepsilon$ -Hermitian space. We denote by $i : G(U) \times G(U) \rightarrow G(W)$ the obvious embedding.

Fix the following datum: $\psi : F \backslash \mathbb{A} \rightarrow \mathbb{C}^\times$ a non-trivial additive character. $\chi_U, \chi_V : E^\times \backslash \mathbb{A}_E^\times \rightarrow \mathbb{C}^\times$, two characters on $\mathbb{A}_E^\times$ such that $\chi_U|_{\mathbb{A}^\times} = \eta^n, \chi_V|_{\mathbb{A}^\times} = \eta^m$, where η denotes the quadratic character associated to E/F . The triple (ψ, χ_U, χ_V) determines a splitting $s := s_{\psi, \chi_U, \chi_V} : G(U)(\mathbb{A}) \times$

$H(V)(\mathbb{A}) \rightarrow \text{Mp}(\mathbb{A})$. Pick a Lagrangian of $U \otimes_E V$. Then define the Weil representation: $\omega := \omega_{\psi, \chi_U, \chi_V, X}$ of $G(U)(\mathbb{A}) \times H(V)(\mathbb{A})$ acting on $\mathcal{S}(X(\mathbb{A}))$. We can then define global theta lifting based on it.

Let π be a cuspidal automorphic representation of $G(U)$. Question: when does the theta lifting $\Theta(\pi)$ to $H(V)$ non-vanishing? If non-vanishing, how to characterize it?

Method: assuming $\Theta(\pi) \subset \mathcal{A}_{\text{disc}}(G)$. $\Theta(\pi) \neq 0 \iff \langle \cdot, \cdot \rangle_{\text{Pet}}|_{\Theta(\pi)} \neq 0$. We formally compute the inner product as follows: let $\varphi_1, \varphi_2 \in \pi, \phi_1, \phi_2 \in \omega = \mathcal{S}(X(\mathbb{A}))$.

$$\begin{aligned}
\langle \theta(\varphi_1, \phi_1), \theta(\varphi_2, \phi_2) \rangle_{\text{Pet}} &= \int_{[H]} \left(\int_{[G]} \overline{\varphi_1(g)} \theta(g, h, \phi_1) dg \cdot \overline{\int_{[G]} \varphi_2(g) \theta(g, h, \phi_2) dg} \right) dh \\
&= \int_{[H]} \int_{[G]} \int_{[G]} \overline{\varphi_1(g_1)} \varphi_2(g_2) \overline{\theta(g_1, h, \phi_1)} \theta(g_2, h, \phi_2) dg_1 dg_2 dh \\
(\text{Change models}) &= \int_{[H]} \int_{[G]} \int_{[G]} \overline{\varphi_1(g_1)} \varphi_2(g_2) \theta(i(g_1, g_2), h, \phi) \chi_V(g_2)^{-1} dg_1 dg_2 dh \\
&= \int_{[G]} \int_{[G]} \int_{[H]} \overline{\varphi_1(g_1)} \varphi_2(g_2) \theta(i(g_1, g_2), h, \phi) \chi_V(g_2)^{-1} dg_1 dg_2 dh \\
(\text{Siegel-Weil}) &= \kappa \int_{[G]} \int_{[G]} \overline{\varphi_1(g_1)} \varphi_2(g_2) E(i(g_1, g_2), \Phi_\phi, s) dg_1 dg_2 \\
(\text{Doubling method}) &= \kappa \cdot L(s_0 + \frac{1}{2}, \pi \times \chi_V) \cdot Z^\natural(s_0, \Phi_\phi, \varphi_1, \varphi_2)
\end{aligned}$$

The Siegel-Weil formula has been discussed thoroughly during the last talk. We concentrate here on the two remaining steps and some convergence issues.

2 Change of models

We explain here why the term $\chi_V^{-1}(g_2)$ appears. Let us temporarily denote by W a symplectic space. The global Weil representation is the representation of $\text{Mp}(\mathbb{A})$ on $\mathcal{S}(X(\mathbb{A}))$, where $W = X \oplus Y$ is a polarization of W . Let $W = X' \oplus Y'$ be another polarization. Then there is an isomorphism $\mathcal{F}_{X, X'} : \mathcal{S}(X(\mathbb{A})) \rightarrow \mathcal{S}(X'(\mathbb{A}))$ given by the partial Fourier transform: write $X = X_1 \oplus Z$, $X' = X' \oplus Z$ where $X_1 \cap X'_1 = \{0\}$. Then

$$\mathcal{F}_{X, X'} \varphi(x'_1 + z) = \int_{X_1(\mathbb{A})} \varphi(x_1 + z) \psi(\langle x_1, x'_1 \rangle) dx_1.$$

By Poisson summation formula, we have

$$\sum_{x \in X(F)} \varphi(x) = \sum_{x' \in X'(F)} \mathcal{F}_{X, X'} \varphi(x').$$

Therefore,

$$\theta(g, \varphi) = \theta(g, \mathcal{F}_{X, X'} \varphi).$$

We hence see that the set of “theta functions” is independent of the choice of the polarization. In particular, one can define global theta correspondence based on any polarization.

We also remark that there is “block compatibility” for the Weil representation. Let W_1, W_2 be two symplectic space over F . Fix a polarization $W_1 = X_1 \oplus Y_1$, $W_2 = X_2 \oplus Y_2$. We then have associated Weil representation ω_1 (resp. ω_2) of $\mathrm{Mp}(W_1)(\mathbb{A})$ (resp. $\mathrm{Mp}(W_2)(\mathbb{A})$) on $\mathcal{S}(X_1(\mathbb{A}))$ (resp. $\mathcal{S}(X_2(\mathbb{A}))$).

Let $W = W_1 \oplus W_2$ be the direct sum, which is naturally a symplectic space. The natural embedding $\mathrm{Sp}(W_1)(\mathbb{A}) \times \mathrm{Sp}(W_2)(\mathbb{A}) \rightarrow \mathrm{Sp}(W)(\mathbb{A})$ lifts to a map

$$\iota : \mathrm{Mp}(W_1)(\mathbb{A}) \rightarrow \mathrm{Mp}(W_2)(\mathbb{A}) \rightarrow \mathrm{Mp}(W)(\mathbb{A}).$$

Let ω be the Weil representation of $\mathrm{Mp}(W)(\mathbb{A})$ associated to the polarization $W = (X_1 + X_2) \oplus (Y_1 + Y_2)$. Then the map $\phi_1 \otimes \phi_2 \mapsto \phi_1 \otimes \phi_2$ induces an isomorphism

$$\omega_1 \otimes \omega_2 \cong \omega \circ \iota$$

as representation of $\mathrm{Mp}(W_1)(\mathbb{A}) \times \mathrm{Mp}(W_2)(\mathbb{A})$. As a consequence, for any $g_1 \in \mathrm{Mp}(W_1)(\mathbb{A}), g_2 \in \mathrm{Mp}(W_2)(\mathbb{A})$ and $\phi_1 \in \mathcal{S}(X_1(\mathbb{A})), \phi_2 \in \mathcal{S}(X_2(\mathbb{A}))$, we have

$$\theta(g_1, \phi_1)\theta(g_2, \phi_2) = \theta(\iota(g_1, g_2), \phi_1 \otimes \phi_2).$$

Now back to our notations in the setup. By the result of Harris-Kudla-Sweet, there is an isomorphism $\alpha : \mathcal{S}(X(\mathbb{A})) \cong \mathcal{S}(X(\mathbb{A}))$ such that for any $\phi \in \mathcal{S}(X(\mathbb{A}))$ and $g \in G(\mathbb{A})$, we have

$$\overline{\omega_{\chi_U, \chi_V, \psi}(g)\phi} = \omega_{\chi_U, \chi_V, \psi}(\alpha(\phi))(g^-)\chi_V^{-1}(g), \quad (1)$$

where g^- is the element g but regarded as an element of $G(U^-)$. Note that although $G(U)$ and $G(U^-)$ are canonically identified. The sections

$$s_{\chi_U, \chi_V, \psi} : G(U)(\mathbb{A}) \rightarrow \mathrm{Mp}(\mathbb{A})$$

and

$$s_{\chi_U, \chi_V, \psi} : G(U^-)(\mathbb{A}) \rightarrow \mathrm{Mp}(\mathbb{A})$$

are different.

Summing the equality (1) over $x \in X(F)$, we see that

$$\overline{\theta(g, \phi)} = \theta_-(g, \alpha(\phi))\chi_V^{-1}(g), \quad g \in G(U)(\mathbb{A})$$

where θ_- denotes the θ -function on $G(U^-)(\mathbb{A})$ through the section $s_{\chi_U, \chi_V, \psi}$. There is also similar equality for h .

Combining with the block compatibility, we prove the “change of model” equality.

3 Doubling L -functions

3.1 Definitions

We first recall some doubling L -function from the previous talk by Julia.

Note that $U^\Delta := \{(u, u) \in W\}$ is a maximal isotropic space of W . Let P be the stabilizer of U^Δ in $G(W)$. Then P is a Siegel parabolic subgroup of $G(W)$. We write $H_P : G(W)(\mathbb{A}) \rightarrow \mathbb{R}$ the Harish-Chandra map: on $P(\mathbb{A})$ is defined by $p \mapsto \log|m|$, where $m \in \mathrm{GL}_n(\mathbb{A}_E)$ is the Levi part of p , and the map H_P extends to $G(W)(\mathbb{A})$ via Iwasawa decomposition.

Let $f \in \text{Ind}_{P(\mathbb{A})}^{G(W)(\mathbb{A})} \chi$. We write f_s be the section $f \cdot e^{sH_P(g)} \in \text{Ind}_{P(\mathbb{A})}^{G(W)(\mathbb{A})} \chi \cdot |\cdot|^s$. We can form the Eisenstein series $E(g, f, s)$:

$$E(g, f, s) = \sum_{\gamma \in P(F) \backslash G(W)(F)} f_s(\gamma g), \quad g \in [G(W)], s \in \mathbb{C}$$

It is convergent when $\text{Re}(s)$ is sufficiently large and has a meromorphic continuation to $\mathbb{C}$. It can also be regarded as a meromorphic map from $\mathbb{C} \rightarrow \mathcal{T}([G(W)])$.

Let π be a cuspidal automorphic representation of $G(U)$ and let $\chi : \mathbb{A}_E^\times \rightarrow \mathbb{C}^\times$ be an idèle class character. Let $\varphi_1 \in \pi, \varphi_2 \in \pi^\vee, f \in \text{Ind}_{P(\mathbb{A})}^{G(W)(\mathbb{A})} \chi$. The following integral is called the doubling integral:

$$Z(\varphi_1, \varphi_2, f, s) = \int_{[G(U)] \times [G(U)]} \varphi_1(g_1) \varphi_2(g_2) E(i(g_1, g_2), f, s) \chi(g_2)^{-1} dg_1 dg_2.$$

We first remark on basic analytic properties of the integral:

- (1) Convergence: $\mathcal{A}_{\text{cusp}}(G) \subset \mathcal{S}([G])$. Therefore, for any N , there exists $C = C_N > 0$ such that

$$|\varphi_1(g_1)| \leq C \|g_1\|_G^{-N}, \quad |\varphi_2(g_2)| \leq C \|g_2\|_G^{-N}.$$

The remaining terms are of moderate growth:

$$|E(i(g_1, g_2), f, s)| \leq C \|g_2\|_G^{N_0}, |\chi_2(g_2)| \leq C \|g_2\|_G^{N_1}.$$

- (2) Meromorphic properties: the integral can be regarded as the composition of

$$\mathbb{C} \rightarrow \mathcal{T}([G \times G]), \quad s \mapsto E(i(g_1, g_2), f, s)$$

and

$$\mathcal{T}([G \times G]) \rightarrow \mathbb{C}, \quad \varphi \mapsto \int_{[G \times G]} \varphi_1(g) \varphi_2(g) \varphi(g_1, g_2) \chi(g_2)^{-1} dg_1 dg_2,$$

the first map is meromorphic and the second map is continuous and linear, hence their composition is meromorphic.

3.2 Unfolding

For simplicity, we write (g_1, g_2) for $i(g_1, g_2)$. For $\text{Re}(s)$ large enough, the integral $Z(\varphi_1, \varphi_2, f, \chi, s)$ can be written as

$$\int_{[G] \times [G]} \varphi_1(g_1) \varphi_2(g_2) \left(\sum_{\gamma \in P(F) \backslash G(W)(F)} f_s(\gamma \cdot i(g_1, g_2)) \right) \chi(g_2)^{-1} dg_1 dg_2.$$

We remark that, since the estimate

$$\sum_{\gamma \in P(F) \backslash G(W)(F)} |f_s(\gamma \cdot i(g_1, g_2))| \leq C \|g_1\|^N \|g_2\|^N$$

holds, so we can regroup the sum in any way we like (dominated convergence theorem). The integral then can be written as

$$\int_{[G] \times [G]} \varphi_1(g_1) \varphi_2(g_2) \left(\sum_{\delta \in P(F) \backslash G(W)(F)/G(F) \times G(F)} \sum_{\gamma \in P(F) \cdot \delta \cdot (G(F) \times G(F))} f_s(\gamma \cdot (g_1, g_2)) \right) \chi(g_2)^{-1} dg_1 dg_2,$$

which is the same as

$$\sum_{\gamma \in P(F) \backslash G(W)(F)/G(F) \times G(F)} \int_{(G \times G)^\gamma(F) \backslash G(\mathbb{A}) \times G(\mathbb{A})} \varphi_1(g_1) \varphi_2(g_2) f_s(\gamma \cdot (g_1, g_2)) \chi(g_2)^{-1} dg_1 dg_2,$$

where

$$(G \times G)^\gamma = \text{Stab}_{G \times G}(P\gamma) = \{g \in G \times G \mid P\gamma = P\gamma g\} = \{g \in G \times G \mid \gamma g \gamma^{-1} \in P\}$$

We separate two cases:

- (1) If $\gamma = 1$, then $(G \times G)^\gamma = \{(g_1, g_2) \in G \times G \mid (g_1, g_2) \in P\}$. Recall that P is stabilizer of $U^\Delta \subset W$. One then check directly that $P \cap (G \times G) = G^\Delta$.
- (2) If $\gamma \neq 1$. By some linear algebra computation (see the book of Piatetski-Shapiro-Rallis). There exists a normal subgroup $N \subset (G \times G)^\gamma$ such that $N = N_1 \times N_2$, where N_1 and N_2 are unipotent subgroup of $G_1 \times G_2$ and at least one of N_1 and N_2 are non-trivial. From this, we see that

$$\int_{(G \times G)^\gamma(F) \backslash G(\mathbb{A}) \times G(\mathbb{A})} f_s(\gamma \cdot (g_1, g_2)) = 0$$

by cuspidality conditions on φ_1 and φ , this vanishes.

Therefore

$$Z(\varphi_1, \varphi_2, f, \chi, s) = \int_{G^\Delta(F) \backslash G(\mathbb{A}) \times G(\mathbb{A})} \varphi_1(g_1) \varphi_2(g_2) f_s(g_1, g_2) \chi(g_2)^{-1} dg_1 dg_2.$$

One then check that for $g \in G^\Delta(\mathbb{A})$,

$$f_s(g(g_1, g_2)) = \chi(g) f_s(g_1, g_2),$$

since under the subspace decomposition $W = U^\Delta \oplus U^\nabla$, $g \in G^\Delta(\mathbb{A})$ is given by the matrix $\begin{pmatrix} g & \\ & g \end{pmatrix}$.

Therefore

$$\begin{aligned} Z(\varphi_1, \varphi_2, f, \chi, s) &= \int_{G^\Delta(\mathbb{A}) \backslash G(\mathbb{A}) \times G(\mathbb{A})} \left(\int_{[G]} \varphi_1(gg_1) \varphi_2(gg_2) dg \right) f_s(g_1, g_2) \chi(g_2)^{-1} dg_1 dg_2 \\ &= \int_{G^\Delta(\mathbb{A}) \backslash (G \times G)(\mathbb{A})} \langle R(g_1) \varphi, R(g_2) \varphi \rangle_{\text{Pet}} \cdot f_s(g_1, g_2) \chi(g_2)^{-1} dg_1 dg_2. \end{aligned}$$

3.3 Local doubling integral

Inspiring by the global zeta integral above, we introduce the following local zeta integral. In this subsection, we let F be a local field, and E/F be an étale quadratic algebra. We have local analogue of various Hermitian spaces and their unitary groups as in Section 1.

Let $\chi : E^\times \rightarrow \mathbb{C}^\times$ be a character of $E^\times$. Let π be an admissible representation of $G(F) = G(U)(F)$ and we denote by $\pi^\vee$ its contragradient. Let $f \in \text{Ind}_{P(F)}^{G(W)(F)} \chi$. For $v \in \pi, v^\vee \in \pi^\vee$. We consider the following local doubling zeta integral

$$Z(v, v^\vee, f, \chi, s) = \int_{G^\Delta(F) \backslash (G \times G)(F)} \langle \pi(g)v, \pi^\vee(g)v^\vee \rangle f_s(g_1, g_2) \chi(g_2)^{-1} dg_1 dg_2.$$

We remark on its analytic properties (see Lapid-Rallis):

- (1) The integral is convergent when $\text{Re}(s)$ sufficiently large. Reason: if π is supercuspidal, then the integral is compactly supported. In general, study the relation between parabolic induction and zeta integral, to reduce to the supercuspidal case. (Archimedean: supercuspidal = the group is compact modulo center).
- (2) The integral has meromorphic continuation to $\mathbb{C}$. Also reduces to the supercuspidal case.

3.4 Doubling L -function

Unramified computation: if E/F is unramified and the representation π is unramified. Let $v \in \pi^K$ and $v^\vee \in \pi^{\vee, K}$ such that $\langle v^\vee, v \rangle = 1$. Assume that χ is unramified and f_s is the standard spherical section. Then Jianshu Li showed:

$$Z(v, v^\vee, f, \chi, s) = \frac{L(s + \frac{1}{2}, \pi \times \chi)}{\prod_{j=1}^n L(2s + j, \chi|_{F^\times} \cdot \eta_{E/F}^{n+j})},$$

as a consequence: if π is a cuspidal automorphic representation of unitary group over global field, then $L(s, \pi \times \chi)$ has meromorphic continuation!

Lapid-Rallis later defined a doubling L -factor for any irreducible smooth representation π of $G(F)$. Yamana interprete this as a g.c.d. Let us temporarily call this function by $L^D(s, \pi \times \chi)$ (But elsewhere we denote it by $L(s, \pi \times \chi)$ for simplicity).

Conjecture 3.1. For any irreducible representation π of $G(F)$, $L^D(s, \pi \times \chi) = L(s, \text{BC}(\pi) \times \chi)$.

We put the normalized doubling zeta integral as

$$Z^\natural(v, v^\vee, f, \chi, s) = \frac{Z(v, v^\vee, f, \chi, s)}{L(s + \frac{1}{2}, \pi \times \chi)}.$$

4 The case $m = n$

In this case, we have conservation law and theta dichotomy. Therefore, given a cuspidal automorphic representation π of $G(\mathbb{A})$, for each non-Archimedean local place v , there exists exactly one local Hermitian space V_v such that the theta lifting of π_v to V_v is non-zero. Let $\mathbb{V}$ be the (possibly) incoherent Hermitian space formed by these V_v .

- (1) If $\mathbb{V}$ is coherent, equivalently $\varepsilon(\frac{1}{2}, \pi) = 1$ by epsilon dichotomy. Let us denote it by V . Then the non-vanishing of theta lifting is equivalent to non-vanishing of L -function $L(\frac{1}{2}, \pi)$.
- (2) If $\mathbb{V}$ is incoherent, then for any coherent V , the theta lifting to V vanishes. We are then led to consider the central derivative, which is the story of arithmetic theta lifting.